

MH3100 Real Analysis I

Tutorial 4 Solutions

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Overview

- **Lecture coverage:** Lectures 10–12 — series, the Cauchy Criterion for Series, the Alternating Series Test, the Comparison Test, the Absolute Convergence Test, absolute vs. conditional convergence, and rearrangements.
- **Questions:** 8 questions (Questions 1–5, Additional Questions A1–A3).
- **Notation:** $\mathbb{N} = \{1, 2, 3, \dots\}$ ($0 \notin \mathbb{N}$ in this course); series $\sum a_n$ with partial sum $s_n := a_1 + \dots + a_n$; results cited by slide name.
- **Running lemmas:** geometric series, $\sum 1/n^2$, and divergence of $\sum 1/n$ are proved once in Question 1 and reused throughout.

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Question 1 (Convergence of three series)

Problem

Determine if each of the following series is convergent or divergent.

(a) $\sum \frac{1}{2^n + n}.$

(b) $\sum \frac{\sin(n)}{n^2}.$

(c) $1 - \frac{1}{2^2} + \frac{1}{3} - \frac{1}{4^2} + \frac{1}{5} - \frac{1}{6^2} + \frac{1}{7} - \frac{1}{8^2} + \cdots.$

(Tutorial 4, Question 1.)

Official hint. Answers only: (a) convergent; (b) convergent; (c) divergent. No method, no comparison series, and no explanation of the pattern in (c) are given.

Solution

Method 1: Comparison Test

Lemma A. $0 < r < 1 \implies \sum r^n$ converges, with sum $\frac{r}{1-r}.$

Proof. Let $0 < r < 1$ and $s_n := \sum_{k=1}^n r^k$ for $n \in \mathbb{N}.$

$$\begin{aligned} (1-r)s_n &= \sum_{k=1}^n r^k - \sum_{k=2}^{n+1} r^k = r - r^{n+1} \\ \implies s_n &= \frac{r - r^{n+1}}{1-r} \end{aligned}$$

$$0 = \lim r^n \therefore \frac{r}{1-r} = \lim s_n \text{ (ALT).}$$

$$\therefore \sum r^n \text{ converges, with sum } \frac{r}{1-r}.$$

□

Lemma B. $\sum 1/n^2$ converges.

Proof. For every $n \geq 2:$

$$0 < \frac{1}{n^2} \leq \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$$

Let $t_N := \sum_{n=1}^N \frac{1}{n^2}.$ For every $N \geq 2:$

$$t_N \leq 1 + \sum_{n=2}^N \left(\frac{1}{n-1} - \frac{1}{n} \right) = 2 - \frac{1}{N} \leq 2 \quad (\text{telescoping})$$

$\{t_N\}$ is increasing and bounded above by 2.

$\therefore \{t_N\}$ converges (MCT) $\therefore \sum 1/n^2$ converges.

□

Lemma C. $\sum 1/n$ diverges; its partial sums $H_N := \sum_{n=1}^N \frac{1}{n}$ are unbounded.

Proof. Claim: $H_{2^m} \geq 1 + \frac{m}{2}$ for every $m \in \mathbb{N}_0$.

$$m = 0: H_1 = 1 \geq 1.$$

If the claim holds for m : the 2^m terms with $2^m < n \leq 2^{m+1}$ are each $\geq \frac{1}{2^{m+1}}$.

$$H_{2^{m+1}} = H_{2^m} + \sum_{n=2^m+1}^{2^{m+1}} \frac{1}{n} \geq 1 + \frac{m}{2} + 2^m \cdot \frac{1}{2^{m+1}} = 1 + \frac{m+1}{2}$$

$1 + \frac{m}{2}$ is unbounded (Archimedean) $\therefore$ the subsequence $\{H_{2^m}\}$ is unbounded, so $\{H_N\}$ is unbounded.

$\therefore \sum 1/n$ diverges (Boundedness of Convergent Sequences). $\square$

Part (a).

Proof. For every $n \in \mathbb{N}$: $2^n + n > 2^n > 0 \therefore$

$$0 < \frac{1}{2^n + n} < \frac{1}{2^n} = \left(\frac{1}{2}\right)^n$$

$\sum (1/2)^n$ converges (Lemma A, $r = \frac{1}{2}$).

$\therefore \sum \frac{1}{2^n + n}$ converges (Comparison Test). $\square$

(a) $\sum \frac{1}{2^n + n}$ is convergent.

Part (b).

Proof. For every $n \in \mathbb{N}$: $|\sin(n)| \leq 1 \therefore$

$$0 \leq \left| \frac{\sin(n)}{n^2} \right| \leq \frac{1}{n^2}$$

$\sum 1/n^2$ converges (Lemma B).

$\therefore \sum |\sin(n)/n^2|$ converges (Comparison Test).

$\therefore \sum \frac{\sin(n)}{n^2}$ converges (Absolute Convergence Test). $\square$

(b) $\sum \frac{\sin(n)}{n^2}$ is convergent (in fact, absolutely).

Part (c).

Proof. Let $b_k := \frac{1}{2k-1}$, $c_k := \frac{1}{(2k)^2}$ for $k \in \mathbb{N}$ and S_N the partial sums $\therefore a_{2k-1} = b_k$, $a_{2k} = -c_k$. For every $m \in \mathbb{N}$:

$$S_{2m} = \sum_{k=1}^m b_k - \sum_{k=1}^m c_k$$

For every $k \in \mathbb{N}$: $0 \leq c_k \leq \frac{1}{k^2}$.

$\therefore \sum c_k$ converges (Lemma B, Comparison Test).

$\therefore$ there is $C > 0$ with $\sum_{k=1}^m c_k \leq C$ for all $m \in \mathbb{N}$ (Boundedness of Convergent Sequences).

For every $k \in \mathbb{N}$: $2k - 1 < 2k \therefore b_k \geq \frac{1}{2k}$.

$$\begin{aligned} \therefore S_{2m} &\geq \frac{1}{2} \sum_{k=1}^m \frac{1}{k} - C & (m \in \mathbb{N}) \\ &= \frac{1}{2} H_m - C \end{aligned}$$

H_m is unbounded (Lemma C) $\therefore$ for every $K \in \mathbb{R}$ there is $m \in \mathbb{N}$ with $\frac{1}{2} H_m - C > K$
 $\therefore S_{2m} > K$, so $\{S_{2m}\}$ and hence $\{S_N\}$ is unbounded.

$\therefore \{S_N\}$ diverges (Boundedness of Convergent Sequences). $\square$

(c) $1 - \frac{1}{2^2} + \frac{1}{3} - \frac{1}{4^2} + \frac{1}{5} - \frac{1}{6^2} + \cdots$ is divergent.

Question 2 (True/false with examples)

Problem

Determine if each of the following statements is true.

- (a) There are two divergent series $\sum a_n$ and $\sum b_n$ where $\sum a_n b_n$ converges.
- (b) There is a convergent series $\sum a_n$ and a bounded sequence $\{b_n\}$ such that $\sum a_n b_n$ diverges.
- (c) There are two sequences $\{a_n\}$ and $\{b_n\}$ where $\sum a_n$ and $\sum(a_n + b_n)$ both converge but $\sum b_n$ diverges.
- (d) There is a sequence $\{a_n\}$ satisfying $0 \leq a_n \leq \frac{1}{n}$ where $\sum(-1)^n a_n$ diverges.

(Tutorial 4, Question 2; part (a) is also Sample Midterm Question 2(a).)

Official hint. Answers only: (a) True; (b) True; (c) False; (d) True. No justification, examples, or proofs are given.

Solution

Method 1: explicit examples; Algebraic Limit Theorem

Part (a). $a_n = b_n := \frac{1}{n}$.

$$\sum a_n = \sum b_n = \sum \frac{1}{n} \text{ diverges (Lemma C).}$$

$$a_n b_n = \frac{1}{n^2} \text{ for all } n \in \mathbb{N} \therefore \sum a_n b_n \text{ converges (Lemma B).}$$

(a) True, e.g. $a_n = b_n = \frac{1}{n}$.

Part (b). $a_n := \frac{(-1)^n}{n}$, $b_n := (-1)^n$.

$$\{1/n\} \text{ is decreasing with } 0 = \lim \frac{1}{n} \therefore \sum (-1)^{n-1} \frac{1}{n} \text{ converges (AST).}$$

$$\therefore \sum a_n \text{ converges (ALT).}$$

$$|b_n| = 1 \text{ for all } n \therefore \{b_n\} \text{ is bounded.}$$

$$a_n b_n = \frac{((-1)^n)^2}{n} = \frac{1}{n} \quad (n \in \mathbb{N})$$

$$\therefore \sum a_n b_n = \sum \frac{1}{n} \text{ diverges (Lemma C)}$$

(b) True, e.g. $a_n = \frac{(-1)^n}{n}$, $b_n = (-1)^n$.

Part (c).

Proof. Suppose $s = \sum a_n$ and $t = \sum(a_n + b_n)$ converge.

$$b_n = (a_n + b_n) - a_n \quad \text{for all } n \in \mathbb{N}$$

$\therefore \sum b_n$ converges, with sum $t - s$ (ALT for series).

$\therefore \sum b_n$ cannot diverge. □

(c) False: $\sum a_n, \sum(a_n + b_n)$ convergent $\implies \sum b_n$ convergent.

Part (d). $a_n := \frac{1}{n}$ (n even), $a_n := 0$ (n odd) $\therefore 0 \leq a_n \leq \frac{1}{n}$ for all $n \in \mathbb{N}$.

Let $T_N := \sum_{n=1}^N (-1)^n a_n$. For every $m \in \mathbb{N}$:

$$T_{2m} = \sum_{k=1}^m \frac{1}{2k} = \frac{1}{2} H_m \quad (a_n = 0 \text{ for odd } n)$$

$$T_{2m+1} = T_{2m} \quad (a_{2m+1} = 0)$$

H_m is unbounded (Lemma C) $\therefore \{T_N\}$ is unbounded.

$\therefore \sum (-1)^n a_n$ diverges (Boundedness of Convergent Sequences).

(d) True, e.g. $a_n = 1/n$ for even n , $a_n = 0$ for odd n .

Question 3 (Proofs or counterexamples)

Problem

For each of the following statements, provide either a short proof if it is true or a counterexample if it is false.

- (a) If a series $\sum a_n$ converges absolutely, then the series $\sum a_n^2$ also converges absolutely.
- (b) If a series $\sum a_n$ converges and a sequence $\{b_n\}$ converges, then the series $\sum a_n b_n$ converges.
- (c) If a series $\sum a_n$ converges conditionally, then the series $\sum n^2 a_n$ diverges.

(Tutorial 4, Question 3.)

Official hint. (a) Apply the Comparison Test using the Terms of Convergent Series, the Boundedness of Convergent Sequences, and the Algebraic Limit Theorem. (b) Find a conditionally convergent series written as $\sum a_n b_n$ with $\sum a_n$ conditionally convergent and $\{b_n\}$ convergent. (c) Prove the contrapositive, applying the Terms of Convergent Series to $\sum n^2 a_n$, then the Comparison Test with a specific absolutely convergent series. No inequality chains, comparison series, or worked examples are given.

Solution

Method 1: Comparison Test; explicit counterexample

Part (a) – true.

Proof. $\sum |a_n|$ converges $\therefore 0 = \lim |a_n|$ (Terms of Convergent Series).

$\therefore$ there is $M > 0$ with $|a_n| \leq M$ for all $n \in \mathbb{N}$ (Boundedness of Convergent Sequences).

$$0 \leq a_n^2 = |a_n| |a_n| \leq M |a_n| \quad \text{for all } n \in \mathbb{N}$$

$\sum M |a_n|$ converges (ALT for series).

$\therefore \sum a_n^2$ converges (Comparison Test).

$a_n^2 = |a_n^2| \therefore \sum |a_n^2|$ converges, i.e. $\sum a_n^2$ converges absolutely. □

(a) True: $\sum a_n$ absolutely convergent $\implies \sum a_n^2$ absolutely convergent.

Part (b) – false. $a_n = b_n := \frac{(-1)^n}{\sqrt{n}}$.

$\{1/\sqrt{n}\}$ is decreasing with $0 = \lim \frac{1}{\sqrt{n}}$ (Archimedean).

$\therefore \sum a_n$ converges (AST, ALT).

$|b_n| = \frac{1}{\sqrt{n}}$ for all $n \therefore 0 = \lim b_n$.

$$a_n b_n = \frac{((-1)^n)^2}{n} = \frac{1}{n} \quad (n \in \mathbb{N})$$

$$\therefore \sum a_n b_n = \sum \frac{1}{n} \text{ diverges} \quad (\text{Lemma C})$$

(b) False, e.g. $a_n = b_n = \frac{(-1)^n}{\sqrt{n}}$.

Part (c) – true.

Proof. Contrapositive: suppose $\sum n^2 a_n$ converges.

$0 = \lim n^2 a_n$ (Terms of Convergent Series).

$\therefore$ there is $M > 0$ with $|n^2 a_n| \leq M$ for all $n \in \mathbb{N}$ (Boundedness of Convergent Sequences).

$$0 \leq |a_n| \leq \frac{M}{n^2} \quad \text{for all } n \in \mathbb{N}$$

$\sum M/n^2$ converges (Lemma B, ALT for series).

$\therefore \sum |a_n|$ converges (Comparison Test), i.e. $\sum a_n$ converges absolutely, so not conditionally.

$\therefore \sum a_n$ conditionally convergent $\implies \sum n^2 a_n$ diverges. □

(c) True: $\sum a_n$ conditionally convergent $\implies \sum n^2 a_n$ divergent.

Question 4 (Cauchy partial sums without the named tests)

Problem

Prove that if the series $\sum |a_n|$ has a Cauchy sequence of partial sums and the sequence $\{b_n\}$ is bounded, then the series $\sum a_n b_n$ has a Cauchy sequence of partial sums. **Do not use the Cauchy Criterion, the Comparison Test, or the Absolute Convergence Test.** (*Tutorial 4, Question 4.*)

Official hint. (a) Write down the definitions for the sequences of partial sums of both $\sum |a_n|$ and $\sum a_n b_n$ to be Cauchy, and use the boundedness of $\{b_n\}$ to compare them. (b) Follow the proof of the Absolute Convergence Test via the Cauchy Criterion. No explicit inequality or ε -bookkeeping is given.

Solution

Method 1: triangle inequality on partial-sum differences

Proof. Let $M > 0$ with $|b_n| \leq M$ for all $n \in \mathbb{N}$.

$$s_n := \sum_{k=1}^n |a_k|, \quad t_n := \sum_{k=1}^n a_k b_k.$$

Let $\varepsilon > 0$ and $\varepsilon' := \frac{\varepsilon}{M+1}$. $\{s_n\}$ is Cauchy $\therefore$ there is $N \in \mathbb{N}$ with

$$n > m \geq N \implies \sum_{k=m+1}^n |a_k| = s_n - s_m < \varepsilon'$$

For $n > m \geq N$:

$$\begin{aligned} |t_n - t_m| &= \left| \sum_{k=m+1}^n a_k b_k \right| \leq \sum_{k=m+1}^n |a_k| |b_k| && \text{(triangle inequality)} \\ &\leq M \sum_{k=m+1}^n |a_k| = M(s_n - s_m) && (|b_k| \leq M) \\ &< M\varepsilon' = \frac{M}{M+1} \varepsilon < \varepsilon \end{aligned}$$

$\therefore |t_n - t_m| < \varepsilon$ for all $n, m \geq N$ (symmetric in n, m ; trivial if $n = m$).

$\therefore \{t_n\}$ is Cauchy. □

$\sum a_n \text{ Cauchy partial sums, } \{b_n\} \text{ bounded} \implies \sum a_n b_n \text{ has Cauchy partial sums.}$

Question 5 (Regrouping preserves the sum)

Problem

Given a series $\sum a_n$ and a strictly increasing sequence of natural numbers $\{n_k\}$, the regrouping of the terms

$$(a_1 + a_2 + \cdots + a_{n_1}) + (a_{n_1+1} + a_{n_1+2} + \cdots + a_{n_2}) + (a_{n_2+1} + a_{n_2+2} + \cdots + a_{n_3}) + \cdots$$

leads to a series $\sum b_k$ with the terms

$$b_k := a_{n_{k-1}+1} + a_{n_{k-1}+2} + \cdots + a_{n_k} \quad \forall k \in \mathbb{N},$$

where n_0 denotes 0. Prove that if $\sum a_n$ sums to s , then $\sum b_k$ sums to s . (*Tutorial 4, Question 5.*)

Official hint. “Compare the sequence of partial sums of $\sum b_k$ with that of $\sum a_n$.” No further detail is given.

Solution

Method 1: partial sums of $\sum b_k$ form a subsequence

Proof. Let $s_n := \sum_{i=1}^n a_i$ and $\sigma_k := \sum_{j=1}^k b_j$. For every $k \in \mathbb{N}$:

$$\begin{aligned} \sigma_k &= \sum_{j=1}^k \sum_{i=n_{j-1}+1}^{n_j} a_i \\ &= \sum_{i=1}^{n_k} a_i = s_{n_k} \end{aligned} \quad (\text{consecutive disjoint blocks, } n_0 = 0)$$

$\{n_k\}$ is strictly increasing $\therefore \{s_{n_k}\}$ is a subsequence of $\{s_n\}$.

$s = \lim s_n \therefore s = \lim s_{n_k}$ (Subsequential Limits).

$\sigma_k = s_{n_k}$ for all $k \therefore s = \lim \sigma_k$, i.e. $\sum b_k$ converges, with sum s . □

$$\sum a_n \text{ sums to } s \implies \sum b_k \text{ sums to } s.$$

Question A1 (Abel's Test)

Problem

Use the Cauchy Criterion for Series to prove that if the series $\sum a_n$ is convergent and the sequence $\{b_n\}$ is nonnegative and decreasing, then the series $\sum a_n b_n$ is convergent. **Do not use the Monotone Convergence Theorem or the Algebraic Limit Theorem for Series.** [Note: This is known as Abel's Test.] (Tutorial 4, Additional Question 1.)

Official hint. (a) Apply the Cauchy Criterion to $\sum a_n$. (b) Abel's Lemma: for $n, m \in \mathbb{N}$ with $m > n$, $\sum_{k=n+1}^m a_k b_k = s_m b_m - s_n b_{n+1} + \sum_{k=n+1}^{m-1} s_k (b_k - b_{k+1})$, where s_k is the k -th partial sum of $\sum a_n$. (c) "What happens to Abel's Lemma when $\{s_n\}$ is constant?" Neither the proof of Abel's Lemma, the bounding argument, nor the purpose of (c) is given.

Solution

Method 1: Abel's Lemma, Cauchy Criterion

Proof. Let $s_k := \sum_{i=1}^k a_i$. Fix $n \in \mathbb{N}$ and put $d_k := s_k - s_n$ for $k \geq n$.

$\therefore d_n = 0$ and $a_k = d_k - d_{k-1}$ for $k \geq n+1$. For $m > n$:

$$\begin{aligned} \sum_{k=n+1}^m a_k b_k &= \sum_{k=n+1}^m d_k b_k - \sum_{k=n}^{m-1} d_k b_{k+1} && (a_k = d_k - d_{k-1}) \\ &= d_m b_m - d_n b_{n+1} + \sum_{k=n+1}^{m-1} d_k (b_k - b_{k+1}) \\ &= (s_m - s_n) b_m + \sum_{k=n+1}^{m-1} (s_k - s_n) (b_k - b_{k+1}) && (d_n = 0) \end{aligned}$$

$\{b_n\}$ is decreasing and $b_n \geq 0 \therefore 0 \leq b_n \leq b_1$ for all $n \in \mathbb{N}$.

Let $\varepsilon > 0$ and $\varepsilon' := \frac{\varepsilon}{b_1 + 1}$.

$\sum a_n$ converges $\therefore$ there is $N \in \mathbb{N}$ with $k \geq j \geq N \implies |s_k - s_j| < \varepsilon'$ (Cauchy Criterion).

For $m > n \geq N$ (so $|s_k - s_n| < \varepsilon'$ for $n \leq k \leq m$):

$$\begin{aligned} \left| \sum_{k=n+1}^m a_k b_k \right| &\leq |s_m - s_n| b_m + \sum_{k=n+1}^{m-1} |s_k - s_n| (b_k - b_{k+1}) && (b_m \geq 0, b_k - b_{k+1} \geq 0) \\ &\leq \varepsilon' b_m + \varepsilon' (b_{n+1} - b_m) = \varepsilon' b_{n+1} && (\text{telescoping}) \\ &\leq \varepsilon' b_1 < \varepsilon \end{aligned}$$

$\therefore$ for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ with $m > n \geq N \implies \left| \sum_{k=n+1}^m a_k b_k \right| < \varepsilon$ (the case $m = n$ is trivial).

$\therefore \sum a_n b_n$ converges (Cauchy Criterion for Series). □

$$\sum a_n \text{ convergent, } \{b_n\} \text{ nonnegative and decreasing } \implies \sum a_n b_n \text{ convergent.}$$

Question A2 (Infinite products)

Problem

A close relative of the infinite series is the *infinite product*, which is a formal expression of the form

$$\prod b_n := \prod_{n=1}^{\infty} b_n := b_1 b_2 b_3 \cdots,$$

where $\{b_n\}$ is a sequence of real numbers. This infinite product is understood in terms of its sequence of *partial products* $\{p_n\}$, where, for each $n \in \mathbb{N}$, $p_n := b_1 b_2 b_3 \cdots b_n$. An infinite product is said to be *convergent* if its sequence of partial products is convergent.

- (a) Use the Cauchy Criterion to prove that if $\{b_n\}$ and $\{c_n\}$ are sequences for which $1 \leq b_n \leq c_n$ for all $n \in \mathbb{N}$ and $\prod c_n$ is convergent, then $\prod b_n$ is also convergent. **Do not use the Monotone Convergence Theorem.**

For the rest of this question, we consider the special case of infinite products of the form

$$\prod (1 + a_n) = (1 + a_1)(1 + a_2)(1 + a_3) \cdots,$$

where the sequence $\{a_n\}$ is nonnegative.

- (b) Find an explicit formula for the partial products in the case where $a_n = \frac{1}{n}$. Is the infinite product convergent?
- (c) Use the Cauchy Criterion to prove that the infinite product $\prod (1 + a_n)$ is convergent if and only if the series $\sum a_n$ is convergent. **Do not use the Monotone Convergence Theorem.**

(Tutorial 4, Additional Question 2.)

Official hint. (a) Follow the proof of the Comparison Test, using the Cauchy Criterion for sequences (not series). (b) Each partial product is a natural number; the infinite product is not convergent. (c) “If”: use $1+x \leq e^x$. “Only if”: use $(1+x)(1+y) \geq 1+x+y$. None of the actual estimates, the explicit formula, or the unboundedness argument in (b) is carried out.

Solution

Method 1: Cauchy Criterion on partial products

Part (a).

Proof. Let $p_n := b_1 \cdots b_n$ and $q_n := c_1 \cdots c_n$.

$$1 \leq b_k \leq c_k \text{ for all } k \therefore 1 \leq p_m \leq q_m \text{ for all } m \in \mathbb{N}.$$

$$\text{For } n > m: 1 \leq b_{m+1} \cdots b_n \leq c_{m+1} \cdots c_n.$$

$$\begin{aligned} \therefore 0 \leq p_n - p_m &= p_m (b_{m+1} \cdots b_n - 1) && (p_n = p_m b_{m+1} \cdots b_n) \\ &\leq q_m (b_{m+1} \cdots b_n - 1) && (p_m \leq q_m) \\ &\leq q_m (c_{m+1} \cdots c_n - 1) = q_n - q_m \end{aligned}$$

$$\therefore |p_n - p_m| \leq |q_n - q_m| \text{ for all } n, m \in \mathbb{N}.$$

$\{q_n\}$ converges $\therefore$ Cauchy (Cauchy Criterion).

Let $\varepsilon > 0$: there is $N \in \mathbb{N}$ with $|q_n - q_m| < \varepsilon$ for all $n, m \geq N \therefore |p_n - p_m| < \varepsilon$ for all $n, m \geq N$.

$\therefore |p_n - p_m| < \varepsilon$ for all $n, m \geq N$ (symmetric in n, m) $\therefore \{p_n\}$ is Cauchy $\therefore$ converges (Cauchy Criterion). $\square$

$$(a) \ 1 \leq b_n \leq c_n \ \forall n, \prod c_n \text{ convergent} \implies \prod b_n \text{ convergent.}$$

Part (b). For every $n \in \mathbb{N}$:

$$p_n = \prod_{k=1}^n \frac{k+1}{k} = \frac{2}{1} \cdot \frac{3}{2} \cdots \frac{n+1}{n} = n+1 \quad (\text{telescoping})$$

For $K \in \mathbb{R}$ take $n \in \mathbb{N}$ with $n > K$ (Archimedean) $\therefore p_n > K$.

$\therefore \{p_n\}$ is unbounded $\therefore$ divergent (Boundedness of Convergent Sequences).

$$(b) \ p_n = n+1 \text{ for all } n \in \mathbb{N}; \text{ the infinite product } \prod (1 + 1/n) \text{ diverges } (p_n \rightarrow \infty).$$

Part (c). Notation: $s_n := \sum_{k=1}^n a_k$, $p_n := \prod_{k=1}^n (1 + a_k)$.

$a_k \geq 0$ for all $k \therefore p_n \geq 1$ and $\{p_n\}$ is increasing.

Proof (“if”). Assume $\sum a_n$ converges.

Step 1. For $r \in \mathbb{N}$ and $x_1, \dots, x_r \geq 0$ with $S_r := \sum_{i=1}^r x_i < 1$, claim:

$$\prod_{i=1}^r (1 + x_i) \leq \frac{1}{1 - S_r}$$

Induction on r . $r = 1$: $(1 + x_1)(1 - x_1) = 1 - x_1^2 \leq 1$ and $1 - x_1 > 0$.

$r \rightarrow r+1$ (with $S_{r+1} < 1$):

$$\begin{aligned} (1 - S_r) - (1 + x_{r+1})(1 - S_{r+1}) &= x_{r+1}S_{r+1} \geq 0 \\ \implies \prod_{i=1}^{r+1} (1 + x_i) &\leq \frac{1 + x_{r+1}}{1 - S_r} \leq \frac{1}{1 - S_{r+1}} \quad (\text{hypothesis}) \end{aligned}$$

Step 2. $\sum a_n$ converges $\therefore$ there is $N_0 \in \mathbb{N}$ with $0 \leq s_k - s_j < \frac{1}{2}$ for all $k > j \geq N_0$ (Cauchy Criterion for Series).

For $n > m \geq N_0$, Step 1 with $x_i := a_{m+i}$, $r = n - m$:

$$\frac{p_n}{p_m} = \prod_{k=m+1}^n (1 + a_k) \leq \frac{1}{1 - (s_n - s_m)}$$

$m = N_0$: $s_n - s_{N_0} < \frac{1}{2} \therefore \frac{p_n}{p_{N_0}} \leq \frac{1}{1 - (s_n - s_{N_0})} < 2 \therefore p_n \leq 2p_{N_0} =: K$ for all $n \geq N_0$.
 $\therefore$ for $n > m \geq N_0$:

$$\begin{aligned} 0 \leq p_n - p_m &= p_m \left(\frac{p_n}{p_m} - 1 \right) \\ &\leq K \frac{s_n - s_m}{1 - (s_n - s_m)} \\ &\leq 2K(s_n - s_m) \quad (1 - (s_n - s_m) > \tfrac{1}{2}) \end{aligned}$$

Step 3. Let $\varepsilon > 0$. There is $N_1 \geq N_0$ with $0 \leq s_n - s_m < \frac{\varepsilon}{2K+1}$ for all $n > m \geq N_1$ (Cauchy Criterion for Series).

$$\therefore 0 \leq p_n - p_m < 2K \cdot \frac{\varepsilon}{2K+1} < \varepsilon \text{ for all } n > m \geq N_1.$$

$\therefore |p_n - p_m| < \varepsilon$ for all $n, m \geq N_1$ (symmetric in n, m) $\therefore \{p_n\}$ is Cauchy $\therefore$ converges (Cauchy Criterion). $\square$

Proof (“only if”). Assume $\{p_n\}$ converges.

For $x_1, \dots, x_r \geq 0$: $\prod_{i=1}^r (1 + x_i) \geq 1 + \sum_{i=1}^r x_i$ (induction: $\prod_{i=1}^{r+1} (1 + x_i) \geq (1 + \sum_{i=1}^r x_i)(1 + x_{r+1}) \geq 1 + \sum_{i=1}^{r+1} x_i$ by $(1+x)(1+y) \geq 1+x+y$, $x, y \geq 0$).

$$\therefore \text{for } n > m: \frac{p_n}{p_m} = \prod_{k=m+1}^n (1 + a_k) \geq 1 + (s_n - s_m).$$

$$0 \leq s_n - s_m \leq \frac{p_n - p_m}{p_m} \leq p_n - p_m \quad (p_m \geq 1)$$

$\{p_n\}$ converges $\therefore$ Cauchy (Cauchy Criterion).

Let $\varepsilon > 0$: there is $N \in \mathbb{N}$ with $|p_n - p_m| < \varepsilon$ for all $n > m \geq N$ $\therefore |s_n - s_m| < \varepsilon$ for all $n > m \geq N$.

$\therefore |s_n - s_m| < \varepsilon$ for all $n, m \geq N$ (symmetric in n, m) $\therefore \{s_n\}$ is Cauchy $\therefore \sum a_n$ converges (Cauchy Criterion for Series). $\square$

$$(c) \prod (1 + a_n) \text{ convergent} \iff \sum a_n \text{ convergent} \quad (a_n \geq 0).$$

Question A3 (Alternating Series Test via NIP)

Problem

The Alternating Series Test can be proved directly from the Nested Interval Property.

- (a) Given an alternating series $b_1 - b_2 + b_3 - b_4 + \dots$ with the sequence $\{b_n\}$ decreasing and converging to 0, construct a nested sequence of closed and bounded intervals $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$, whose lengths converge to 0, such that for each $n, m \in \mathbb{N}$, $m \geq n \implies s_m \in I_n$, where s_m denotes the m -th partial sum of the alternating series.

- (b) Hence use the Nested Interval Property to deduce a sum for the alternating series.

(Tutorial 4, Additional Question 3.)

Official hint. (a) Recall that the lecture proof of the Alternating Series Test establishes that for every $N, n \in \mathbb{N}$, $n > N \implies 0 \leq b_{N+1} - b_{N+2} + \dots + (-1)^{n-N-1}b_n \leq b_{N+1}$. (b) The Nested Interval Property gives a real number s ; show s is the sum via the definition of a limit, remembering $\{b_n\} \rightarrow 0$. No explicit interval, no nesting check, and no final ε - N argument are given.

Solution

Method 1: nested intervals from consecutive partial sums

Part (a). $s_0 := 0$ and, for $n \in \mathbb{N}$, $I_n := [\min(s_{n-1}, s_n), \max(s_{n-1}, s_n)]$ (closed and bounded).

Containment. Fix $n \in \mathbb{N}$ and $m \geq n$. For $n \geq 2$: the lecture inequality with $N := n - 1 \in \mathbb{N}$:

$$0 \leq b_n - b_{n+1} + \dots + (-1)^{m-n}b_m \leq b_n$$

$(-1)^{k-1} = (-1)^{n-1}(-1)^{k-n} \therefore$ the middle term is $(-1)^{n-1}(s_m - s_{n-1})$.

$(-1)^{n-1}(s_n - s_{n-1}) = b_n \therefore 0 \leq (-1)^{n-1}(s_m - s_{n-1}) \leq (-1)^{n-1}(s_n - s_{n-1})$.

Multiplying by $(-1)^{n-1} = \pm 1$: $s_m - s_{n-1}$ lies between 0 and $s_n - s_{n-1}$.

$\therefore s_m$ lies between s_{n-1} and s_n , i.e. $s_m \in I_n$.

For $n = 1$: $s_0 = 0$, $s_1 = b_1$ and $s_m = b_1 - (b_2 - b_3 + \dots + (-1)^{m-2}b_m)$ for $m \geq 2$.

$0 \leq b_2 - b_3 + \dots + (-1)^{m-2}b_m \leq b_2 \leq b_1$ (lecture inequality, $N = 1$) $\therefore 0 \leq s_m \leq b_1$, i.e. $s_m \in I_1$ for $m \geq 2$; $s_1 = b_1 \in I_1$.

Nesting. $s_n, s_{n+1} \in I_n$ ($m = n, n + 1$ above).

I_{n+1} is the smallest closed interval containing $s_n, s_{n+1} \therefore I_{n+1} \subseteq I_n$ for all $n \in \mathbb{N}$.

Lengths. $b_n \geq b_k$ for all $k \geq n$ and $0 = \lim b_k \therefore b_n \geq 0$ (OLT).

$\therefore |I_n| = |s_n - s_{n-1}| = b_n$, and $0 = \lim b_n$.

Part (b).

Proof. $\{I_n\}$ is nested, closed and bounded $\therefore$ there is $s \in \mathbb{R}$ with $s \in I_n$ for all $n \in \mathbb{N}$ (NIP).

Unique: if also $s' \in I_n$ for all n , then $0 \leq |s - s'| \leq |I_n| = b_n$ for all n .

$\therefore |s - s'| \leq 0$ (OLT) $\therefore s = s'$.

Let $\varepsilon > 0$; $0 = \lim b_n \therefore$ there is $N \in \mathbb{N}$ with $b_N < \varepsilon$.

For every $m \geq N$: $s_m \in I_N$ (part (a)) and $s \in I_N$ $\therefore$

$$|s_m - s| \leq |I_N| = b_N < \varepsilon$$

$\therefore s = \lim s_m$ (def. of limit), i.e. the series converges, with sum s . □

$$b_1 - b_2 + b_3 - \cdots \text{ converges to the unique } s \in \bigcap_{n=1}^{\infty} I_n.$$

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Method Map

Q.	Core topic	Methods included
1	Convergence of three series	Comparison Test (geometric series, $\sum 1/n^2$); Absolute Convergence Test; unbounded subsequence of partial sums.
2	True/false with examples	Explicit examples; Algebraic Limit Theorem for series.
3	Proof or counterexample	Comparison Test with a global bound; explicit counterexample; contrapositive.
4	Cauchy partial sums	Triangle inequality on partial-sum differences.
5	Regrouping preserves the sum	Subsequence of partial sums; Subsequential Limits.
A1	Abel's Test	Abel's Lemma (summation by parts); Cauchy Criterion for Series.
A2	Infinite products	Cauchy Criterion on partial products; telescoping; product bounds $\prod(1+x_i) \leq \frac{1}{1-\sum x_i}$ and $\prod(1+x_i) \geq 1 + \sum x_i$.
A3	Alternating Series Test via NIP	Nested intervals from consecutive partial sums; Nested Interval Property.

Final Answers

- (a) Convergent. (b) Convergent. (c) Divergent.
- (a) True. (b) True. (c) False. (d) True.
- (a) True. (b) False. (c) True.
- $\sum |a_n|$ Cauchy partial sums, $\{b_n\}$ bounded $\implies \sum a_n b_n$ Cauchy partial sums.
- $\sigma_k = s_{n_k}$, so $\sum b_k$ sums to s .
- A1. $\sum a_n b_n$ converges (Abel's Test).
- A2. (a) $\prod b_n$ converges. (b) $p_n = n + 1$; divergent. (c) $\prod(1 + a_n)$ converges iff $\sum a_n$ converges.
- A3. $I_n = [\min(s_{n-1}, s_n), \max(s_{n-1}, s_n)]$; the series converges to the unique $s \in \bigcap_n I_n$.